

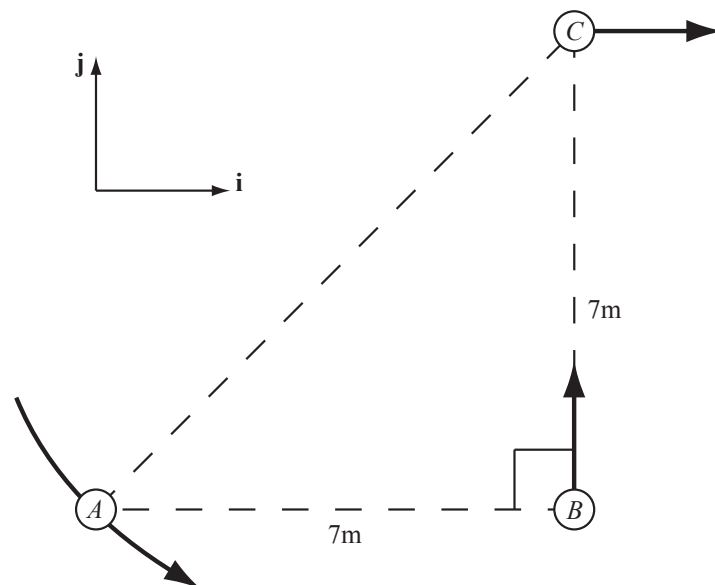
9 (**long**) Three children, A , B and C , are playing a running game, as viewed from above in Fig. 6. At the instant shown, B is 7 m from both A and C , who are approximately 10 m apart. In absolute terms, A is running in a circular path centred on the indicated position of C at 2 m s^{-1} and accelerating along that path at 1 m s^{-2} . In absolute terms, B is running in the $\mathbf{j}$ direction at constant 2 m s^{-1} , and C is running in the $\mathbf{i}$ direction at 2 m s^{-1} and accelerating at 0.5 m s^{-2} in the $\mathbf{i}$ direction.

(a) Write down, in vector terms, the absolute velocities and accelerations of each of the three children separately. [6]

(b) Calculate the velocities of both A and B relative to C . [6]

(c) Calculate the accelerations of both A and B relative to C . [6]

(d) At the instant shown in Fig. 6, A throws a ball such that the velocity of the ball relative to the velocity of A is in the $\overline{AC}$ direction, but the ball is subsequently caught by B . By carefully considering the velocity of the ball, calculate how far B has run by the time she catches the ball. You may neglect both gravity and air resistance and assume that all movement is in the plane shown. [12]



(not to scale)

Fig. 6